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An integrated model for an advanced production process – Agile Re-engineering Project Management

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Abstract: In recent years, the most common need for companies consists in an increasingly efficient business process management and its continuous improvement. Companies are achieving this goal by implementing innovative approaches targeted on the individual project requirements. In this scenario, the present research aims to investigate, through a real pilot project, the benefits of implementing the agile project management (APM) framework based on a multi tools approach in order to re-engineering a business process (BPR). A real case study concerning a manufacturing plant is analyzed, in order to assess and highlight the effectiveness of the proposed model.

Keywords: MCDA, AHP, Simulation, KPI, Business model

1. INTRODUCTION

In the present business context, it is essential to implement continuous innovations in a short period of time. As assed by Alguezaui and Filieri (2010), this condition clashes with the complexity requirements of technological products and leads to increasing difficulties in production processes management. In order to gain a competitive advantage, firms try to implement innovative tools and strategies through large investments in resources and relationships with various partners such as universities, suppliers and customers (Filieri and Alguezaui 2008; De Felice et al. 2018; Cerbaso et al. 2017). One major category of tools used to implement a winning management strategy is known as *project management tools* (Chapman et al., 2004). Many researches have attempted to define what project management is but, maybe the most significant definition is provided by the Project Management Institute "*a temporary group activity designed to produce a unique product, service or result*" (Wisner, 2016). The implementation of traditional project management tools requires that work begins with the elicitation and documentation of a complete set of requirements, followed by high-level design, development, and inspection. However, some practitioners found these initial development steps frustrating and, perhaps, impossible. As a result, several consultants have independently developed methods and practices to respond to the inevitable change they were experiencing. Agile Project Management (APM) is a reaction against traditional methodologies (Hoda et al., 2010; Greco et al. 2016). The concepts of agile product development first emerged among Japanese automobile manufacturers in the 1980s. In particular Gibbs (2006) states that the pioneers of the agile approach followed the example of Toyota's lean production. Agile methods aim to manage the impact of change on a project, breaking down the development of new products/process into a number of repeating cycles called iterations (Aguanno, 2004). To achieve an agile approach, it is necessary to develop adequate strategies to control parameters that affect the quality of the product/process. One of the most popular and well documented change intervention is Business Process Reengineering (BPR), that represents the process improvement of a pre-existing process (Yu, 2001). The implementation of BPR can improve the efficiency and effectiveness of the processes, such as the purchase to pay process, which involves people from a number of different functions (Cheng et al., 2009; Yu, 2011; Glass, et al. 2001). The aim of this paper is to investigate through a pilot project the benefits of the agile project management (APM) framework, thanks to the integration with other management tools, compared to the traditional waterfall model, and to understand how it can help companies in gaining competitive advantage. In particular, the area of action is the analysis and design of business processes. The integrated approach, called Agile Re-engineering Project Management (ARPM), is applied in a real case study. Of course the proposed model is applicable and scalable to any kind of project.
The other part of the paper is organized as follows: in section 2 the rationale of the present research is presented. Section 3 describes the model application. Finally, in section 4, conclusions and research guidelines for future research are summarized.

2. RESEARCH ARCHITECTURE

The aim of the proposed model is the adoption of an agile approach in order to re-engineering a business process, with the adoption of key performance indicators to head the achievement of the project goals. The core of the model is the integration of three main aspects: 1) **Business Process Re-engineering** (BPR): the rethinking of organizational workflows (United States General Accounting Office, 1997);

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2) **Agile Project Management** (APM): the use of short development cycles to focus on continuous improvement (Highsmith, 2004) and 3) **Key Performance Indicators**: the adoption of measurable values to demonstrate the achievement of objectives (Parmenter, 2015).

The advantages of the proposed methodological approach and integration, called Agile Re-engineering Project Management, are: 1) a rapid and flexible deployment of solutions; 2) an optimal and collaborative project control; and 3) a reduction of efforts and a growth of success. In this section the model is presented. The structured research framework in organized in 4 main steps and 10 activities, as shown in Figure 1.

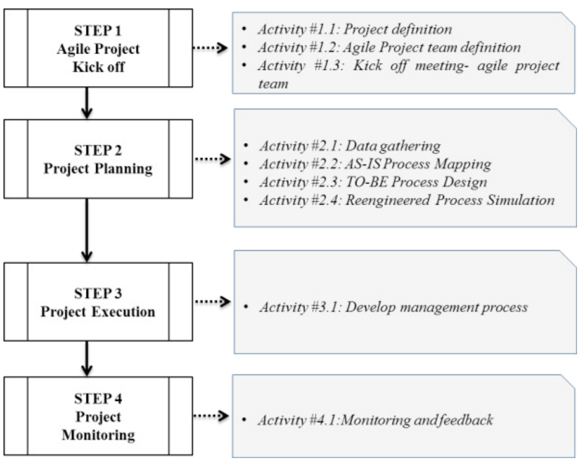

Fig. 1. Summary of phases, steps, activities (author's construction)

A brief description of each steps is provided below.

Step #1. Agile Project Kick off. Preparation of any project is a vital factor for the success of any activities (De Felice et al., 2015; Rossi et al. 2016). It is essential to develop general consensus on the importance of reengineering and the link with agile principles. The main purpose of the present step is: 1) to define and to communicate clearly the requirements; 2) to share responsibility; 3) to embrace the environment of change.

Step #2. Project Planning. The objective of this step is to produce one or more alternatives to the current situation which satisfy the strategic goals of the enterprise. A feasibility assessment basically aims at evaluating the suitability of the project from technical, operational and economic points of view (Di Bona et al., 2016 and Petrillo et al., 2018) a). To achieve this aim, the feasibility assessment encompasses two activities, namely "As Is analysis", "To Be reengineering".

Step #3. Project Execution. The aim of the present step is to align the organizational structure, information systems and the business policies and procedures with the redesigned processes (De Carlo et al. 2013). In order to implement this phase, it is useful to adopt additional requirements such as the Product Backlog, the control chart, etc.

Step #4. Project Monitoring. Many companies build up manufacturing networks to cope with globalized competition. The performance of the process before and after reengineering should be further inspected and validated to ensure the effectiveness of the redesigned process (Romano et al. 2014). Successful implementation of this framework requires a deep understanding and familiarity with the organizational culture and its dynamics and politics (Di Pasquale et al. 2017). The aim of the present step is to develop the proper tools in order to monitor the process such as key performance indicators (KPIs).

3. MODEL APPLICATION

In the present section the model has been applied to a real case study concerning an automotive plant. The different phases, steps and activities depicted in the research framework are analyzed below.

Step #1. Agile Project Kick off

The first step of the research involved the representation of the current layout of the company. The project of re-engineering aimed to move the production line from plant A in Northern Europe towards a more efficient configuration in plant B in Southern Europe.

Activity #1.1: Project definition. The purpose of the project was to increase the company's productivity and profit. The operating parameters were defined as follows: working cycle 8 hours a day: Monday – Friday; production is made in direct distribution based on demand.

Activity #1.2: Agile Project team definition. The teamwork included members from different business functions involved in ARPM model implementation. In detail, the team was composed by 1 Product Owner (PO represents the stakeholders or in other words the customer); 2 production managers, 2 quality managers, 2 technical engineers, 1 plant manager site A, 1 plant manager site B.

The teamwork received specific training regarding the principles characterizing the agile project management and regarding the awareness that agility does not mean that there is no structure.

Activity #1.3: Kick off meeting - agile project team. The project required a formal kick-off meeting to bring together the team and stakeholders. Thus, the kick off meeting for the project was the best opportunity to establish a common purpose toward completing the work. To be an effective activity it was necessary to share with all participants how to work. Furthermore, a tool for mapping the companies' business models was used. In detail, the ***Business Model Canvas (BMC)*** template, which is a visual chart divided into nine building blocks was applied. Figure 2 shows an example used for BMC.

Step #2. Project Planning

Process evaluation analyzed how activities are implemented and machinery is used.

Activity #2.1: Data gathering. During data gathering the following approaches were used: 1) direct interaction with customer; 2) direct interaction with plant managers.

Activity #2.2: AS-IS process mapping and analyzing. The aim of the present activity was to define AS-IS process. For this purpose, the lay-out was analyzed. In detail, the re-engineering of the production line involves the following machines: 6 bending (machines code B1-B2-B3-B4-B5-B6); 2 presses (machines code P1-P2); 2 Neckers; 8 banks for forming.

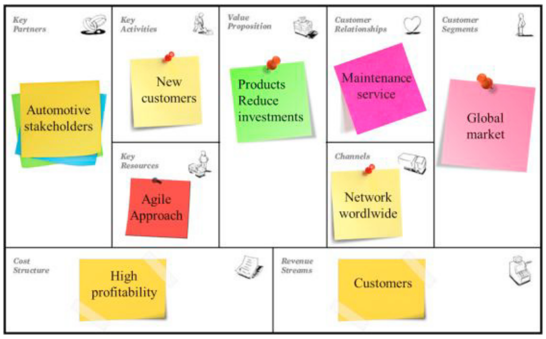

Fig. 2. Example of BMC (team's elaboration)

Table 1 shows some details about the most relevant machines (B and P).

Table 1. Dimensions of the machines (B and P)

Machines	Width (m)	Length (m)	Surface (m2)
B1 (code N200440)	4.95	8.50	42.1
B2 (code N200447)	4.95	8.50	42.1
B3 (code N200187)	2.15	7.00	15.0
B4 (code N200175)	3.35	7.00	23.5
B5 (code N200174)	2.75	7.00	19.3
B6 (code N200441)	2.10	5.40	11.3
P1 (code N200145)	1.50	2.50	3.8
P2 (code N200150)	2.40	2.00	4.8

Activity #2.3: TO-BE process design. In re-engineering a process, many alternatives are often possible and have to be discussed by the agile project team. ARPM suggests the adoption of a multi criteria approach, the ***Analytic Hierarchy Process*** method (Saaty, 1977), able to guide towards the optimal choice in a faster way. Therefore, once defined the current situation, the next step was to analyze the optimal layout for plant B. The main weakness to be solved for the new plant B was to balance the line because workstations work with different workloads. For this purpose, two proposals were analyzed.
In the first hypothesis (Figure 3) production cycles were analyzed in order to reduce the times of production and to facilitate the operator's movements. All machines were positioned along the perimeter of the department, so as to leave a central space for the passage, through which it is ensured the communication between the various areas of the department and the warehouses. Machines area is approximately 620 m^2, with a considerable saving of space compared to the surface they currently occupied, which is equal to about 700 m^2.

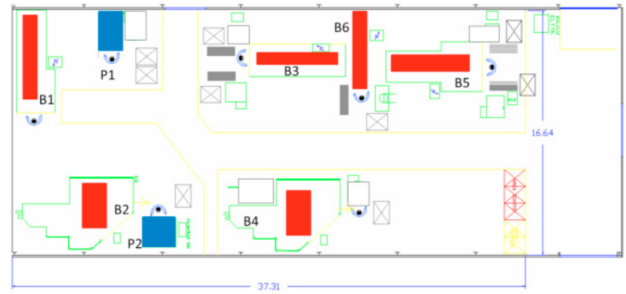

Fig. 3. Proposal N°1. Lay-out plant configuration

In the second proposal (Figure 4) the most relevant difference is related to the position of B3, B6 and B5. Production cycles remain unchanged. The advantage with this hypothesis is therefore attributable to savings in the area of surface occupied by machines, which is reduced to about 600 m^2, compared to 700 m^2 of the current department.

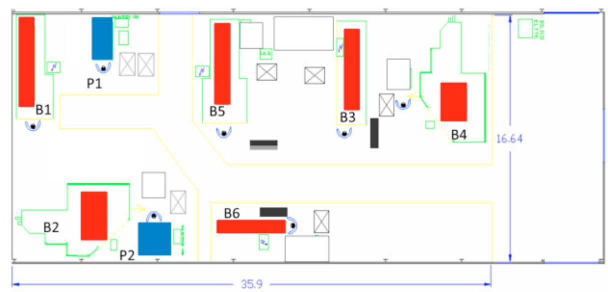

Fig. 4. Proposal N°2. Lay-out plant configuration

The choice of the optimum layout between the two different solutions was made using AHP, that provided to judge each lay-out alternative with reference to the same factor using a numeric value (Di Bona et al., 2016; Di Bona et al. 2018). According to AHP, Saaty's scale was used. Saaty's scale is a semantic scale of 1-9 when comparing two components. For example, number 9 represents the extreme importance over another element. The consistency index (CI) for all matrices of judgment is calculated according to: Consistency Index = CI = (λ_{max} - n) /(n-1) where λ_{max} is the maximum eigenvalue and n is matrix dimension. The parameters considered in the choice of the new plant are: Flexibility (C1); Efficiency (C2); Good use of space (C3); Occupied area m^3 (C4); and Performance/technology (C5). The above parameters have been evaluated, by Agile Project Management Team, for 2 different possible solutions for plant B: alternative α (first hypothesis) and alternative β (first hypothesis). Table 2 summarizes the qualitative and quantitative features for each alternative.

Table 2. Qualitative and Quantitative features for α and β

Criteria	Alternative α	Alternative β
Flexibility (C1)	Good	Low
Efficiency (C2)	60%	70%
Good use of space m^2(C3)	85%	75%
Material handling (C4)	Good	Good
Performance/technology (C5)	Excellence	Good

Table 3. Example of AHP matrix

	C1	C2	C3	C4	C5	Weight
C1	1	4	3	5	3	12.4%
C2	1/4	1	2	1/2	1/3	9.18%
C3	1/3	1/2	1	1/2	1/2	7.91%
C4	1/5	2	2	1	1/2	50.3%
C5	1/3	3	2	2	1	20.10%
CI	**0.050**					
Weight Solution α	**72.6%**					
Weight Solution β	**27.3%**					

In order to define the final rank, AHP Matrix was built (see Table 3). The best layout was the solution α with a total score of 72.6%.

Activity #2.4: Re-engineering Process Simulation

In re-defining the workflows of a process, ***Simulation*** tools become helpful instruments of analysis, to predict the results of the previous choices. At this point, once defined the general layout, it was necessary to design the detailed layout. A simulation model was built in order to verify the optimal layout. Figure 6 shows the first model. The main feature of the present model is that the operator is able to insert and withdraw the products from machinery remained stable in his/her position. It means that it is possible to avoid unnecessary movements and to reduce the total time of the processing cycle.

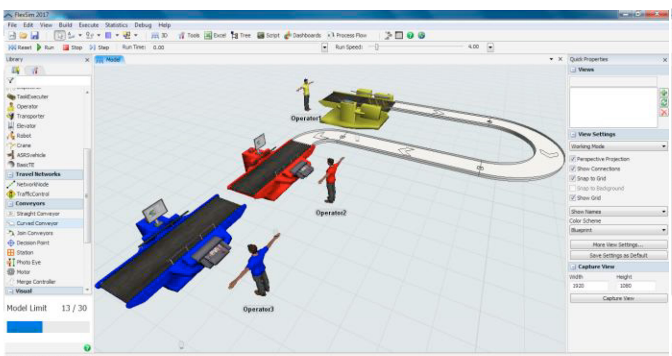

Fig. 5. Simulation model

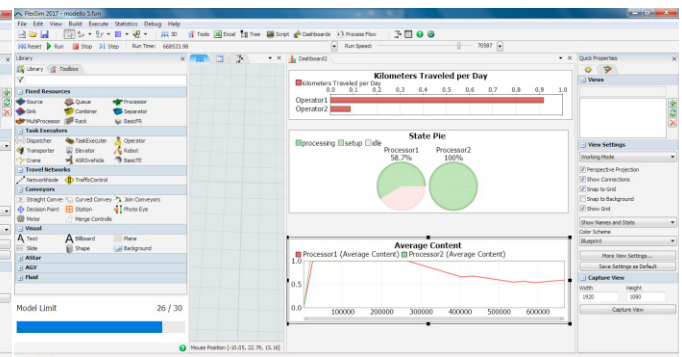

Fig. 6. Simulation model – data report

While, Figure 6 shows an example of data obtained through the simulation.

In detail the most important results are related to B3 and B4:

- for B3 the percentage of time in which the machine is used (% Busy) is 19.72% (before the optimization the percentage was 6.65%);
- for B4 the percentage of time in which the machine is used (% Busy) is 27.55% (before the optimization the percentage was 4.64%)

Furthermore, as shown in Table 4 the new configuration represents the optimal solution, because all machines work.

Results show that the total production time decreases. Considering that each work shift is about € 120.00 it means that an economic saving of about € 2,600.00 monthly is expected. In addition, the new configuration ensures a decrease of processing time for some items. It means that a total savings of about € 2,750.00 monthly is expected. Definitively the offshoring and re-designing of the production line from plant A in Northern Europe to a plant B in Southern Europe does not involve any additional costs, only those related to the relocation of machinery.

Table 4. Numerical machine statistics

	BEFORE Plant A (old)		AFTER Plant B (new)	
Machines	% busy time	Working hours (m)	% busy	Working hours (m)
B1 N200187	6.65%	29.5	19.72%	88.4
B2 N200182	15.99%	70.9	15.81%	70.9
B3 N200184	7.38%	32.7	7.30%	32.7
B4 N200175	4.64%	20.6	27.55%	123.5
B5 N200183	6.33%	28.0	6.26%	28.1
B6 N200181	22.21%	98.4	21.96%	98.4
TOTAL		**280.0**		**441.9**

Step #3. Project Execution.

The aim of the present step was to align the organizational structure, information systems and the business policies and procedures with the redesigned processes. For this purpose, according to agile approach, it was useful to adopt additional requirements such as Product Backlog, burn down chart, progress chart, etc. All work was done in sprints. Each sprint was an iteration of 5 consecutive calendar days. Each sprint was initiated with a sprint initial meeting, where the Product Owner (PO) and Team get together to collaborate to conduct the next Sprint. At the end of the sprint, the PO attended the sprint review meetings and compare the project's actual progress against its planned progress.

Activity #3.1: Process Priority Control. During the course of a project, it was important to define a Product Backlog or a prioritized features list, containing short descriptions of all functionality desired. Furthermore, to track the team's progress in order to define its needs some corrective actions were taken. Thus, a graphical representation (see Figure 7), through a Burn down chart, of work left to do versus time was carried out. The vertical axis points out the outstanding work (or backlog), while the horizontal axis points out the time.

Step #4. Project Monitoring

In order to achieve organizational goals through progress definition and measurement it was useful to define the performance measurement dimensions. From this point of view, specific **Key Performances Indicators** (KPIs) were agreed upon by an organization. These indicators can be used to measure reflecting success factors.

Activity #4.1: Monitoring and feedback. It is possible to define different KPIs for different dimensions. The KPIs selected reflected the organization's goals. KPIs must be measurable. A successful KPI implementation depends on the organization's ability to quickly gather and distribute this key business information. Traditional systems can't provide real time information and require far too much effort and money. A possible set of KPIs is defined in Table 5.

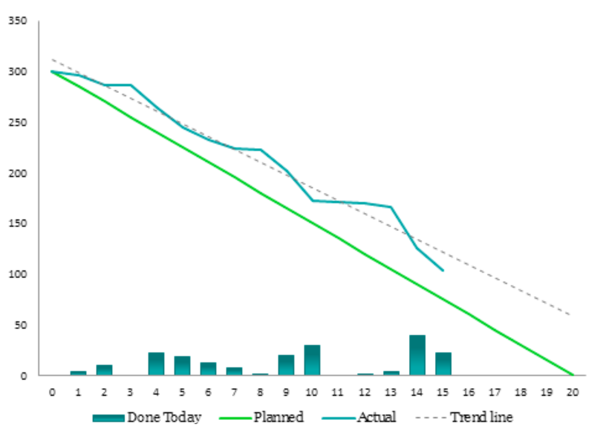

Fig. 7. Example of Burn down chart

Table 5. Example of KPIs

ID		Name	Frequency
8		Production losses	Weekly
7		Production cost variance	Weekly
3		Labour cost per unit	Monthly
4		Availability OEE	Monthly
1		Value of work in progress	Daily
5		Cycle Time Ratio – CTR	Weekly
6		Labour Costs	Monthly

5. DISCUSSION

The common four phases of Agile Project Management: Scan; Analyze; Respond; Change are linked to the common four phases of Business Process Re-engineering: Identify Process; Analyze As-Is; Design To-Be; Test & Implement To-Be; through the integration with other tools: Business Model Canvas; Analytic Hierarchy Process; Simulation; Key Performances Indicators. The result is the definition of a detailed methodology, Agile Re-engineering Project Management, able to integrate qualitative and quantitative techniques. Even if the focus is the analysis and re-design of workflows and business processes, the approach presents general characteristics suitable to any project of continuous improvement. The key factors are: clear definition of objectives; team working valorisation; integration of competences and tools; continuous monitoring of results; quick decisions and changes; high levels of adaptability.

6. CONCLUSIONS

In response to the economic downturn following the recent financial crisis, several companies try to explore the possibility to re-define their processes, sometimes by transferring production in other plants or counties. This paper examines a novel methodological approach based on agile project management. Agile methodologies generally promote a project management process that encourages stakeholder involvement, feedback, objective metrics and effective controls. In general, it is simply to find good practices but no real evidence how to develop and how to implement an agile approach within organizations. The main implications of our work for research and business is to propose a structured methodological approach, rigorous but simple, suitable to implement in any companies. The main strength of the proposed integrated model, ARPM, is to combine qualitative and quantitative evaluations trough a rigorous approach based on APM. The methodology has been investigated thanks to the application to the re-designing and the realization of a production line in consequence of its relocation. The proposed model should serve as a valuable tool to facilitate a successful BPR design in the project management and intends to assist companies as they operate projects of transferring and optimizing production lines. This approach can give full visibility of all processes within a company. This represents a key feature of the model. The novelty of the approach is the development of agile approach outside the field of IT projects as in a manufacturing company. The main limits of the model are: 1) it requires close collaboration among team; 2) it is rather intense for developers; 3) it is necessary flexibility to change course as needed and to ensure delivery of the right product.

The aim of future research is twofold. Firstly, is to investigate a more complex scenario to apply the model with aggressive deadlines, a high degree of complexity and a high degree of uniqueness. Secondly, is to develop several projects in order to be a reference point and a benchmark for several companies in different sectors.

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